Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Ethernet client port

File: eth_client.kicad_sch

Ethernet I/O port

File: eth_io.kicad_sch

Memories and display

File: periph.kicad_sch

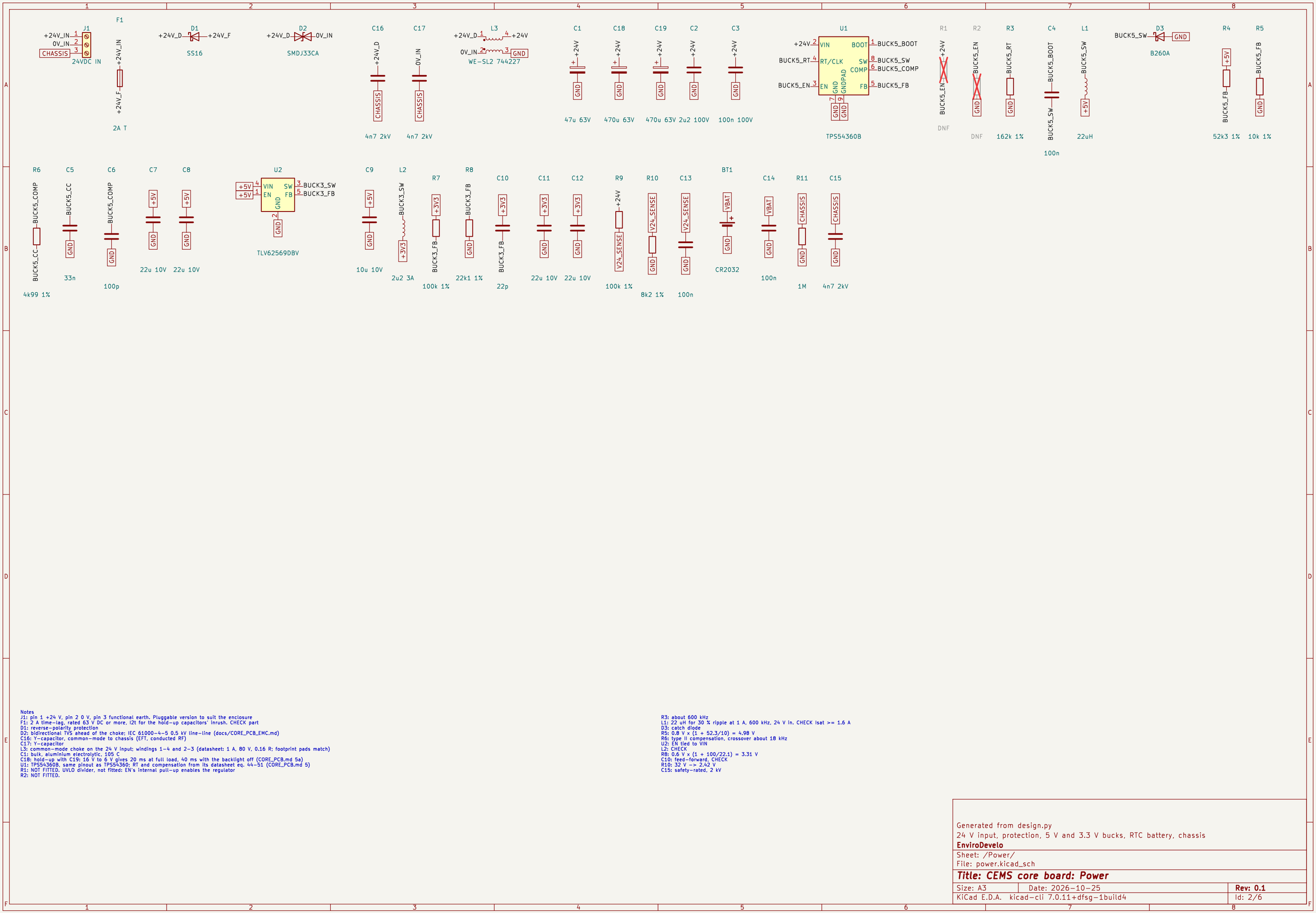
CEMS core board, rev 0.1 schematic (draft for review)
Two separate Ethernet networks: client port = STM32H573 MAC + LAN8742A (J2); I/O port = W5500 (J3), I/O boards only.
24 VDC in; DIN-rail enclosure behind the RVT70 display (FFC J4).
Pin map and design notes: docs/CORE_PCB.md. Parts marked CHECK in the sheet notes need datasheet confirmation before layout.

Design spec: docs/CORE_PCB.md
Generated from hardware/core_pcb/tools/design.py; do not edit by hand
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Sheet: /
File: cems_core.kicad_sch

Title: CEMS core board

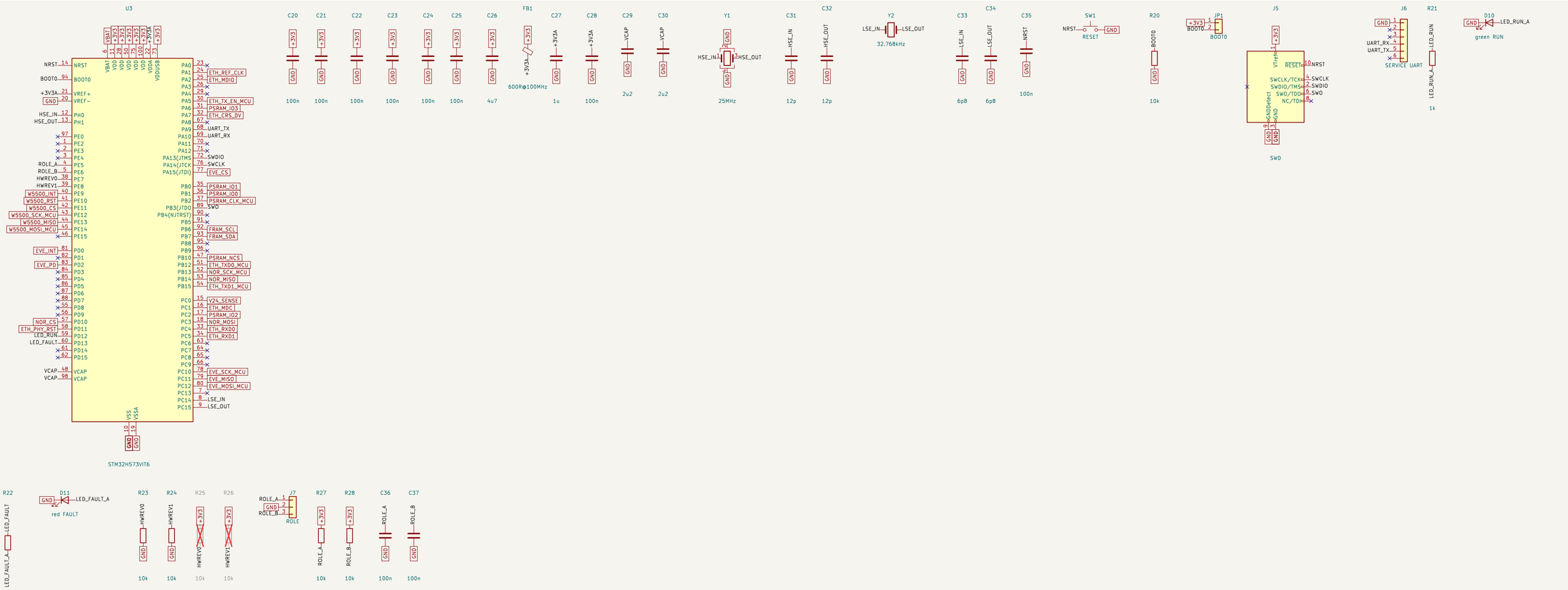
Size: A4	Date: 2026-10-25	**Rev: 0.1**
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Notes
J1: pin 1 +24 V, pin 2 0 V, pin 3 functional earth. Pluggable version to suit the enclosure
F1: 2 A time-lag, rated 63 V DC or more, I2t for the hold-up capacitors' inrush. CHECK part
D1: reverse-polarity protection
D2: bidirectional TVS ahead of the choke; IEC 61000-4-5 0.5 kV line-line (docs/CORE_PCB_EMC.md)
C16: Y-capacitor, common-mode to chassis (EFT, conducted RF)
C17: Y-capacitor
L3: common-mode choke on the 24 V input; windings 1-4 and 2-3 (datasheet: 1 A, 80 V, 0.16 R; footprint pads match)
C1: bulk, aluminium electrolytic, 105 C
C18: hold-up with C19: 16 V to 6 V gives 20 ms at full load, 40 ms with the backlight off (CORE_PCB.md 5a)
U1: TPS54360B, same pinout as TPS54360; RT and compensation from its datasheet eq. 44-51 (CORE_PCB.md 5)
R1: NOT FITTED. UVLO divider, not fitted: EN's internal pull-up enables the regulator
R2: NOT FITTED.
R3: about 600 kHz
L1: 22 uH for 30 % ripple at 1 A, 600 kHz, 24 V in. CHECK Isat >= 1.6 A
D3: catch diode
R5: 0.8 V x (1 + 52.3/10) = 4.98 V
R6: type II compensation, crossover about 18 kHz
U2: EN tied to VIN
L2: CHECK
R8: 0.6 V x (1 + 100/22.1) = 3.31 V
C10: feed-forward, CHECK
R10: 32 V -> 2.42 V
C15: safety-rated, 2 kV

Generated from design.py
24 V input, protection, 5 V and 3.3 V bucks, RTC battery, chassis
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Sheet: /Power/		
File: power.kicad_sch		
Title: CEMS core board: Power		
Size: A3	Date: 2026-10-25	**Rev: 0.1**
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Notes
U3: STM32H563VIT6 is pin-compatible
C20: one per VDD pin, VDDUSB included
C21: one per VDD pin, VDDUSB included
C22: one per VDD pin, VDDUSB included
C23: one per VDD pin, VDDUSB included
C24: one per VDD pin, VDDUSB included
C25: one per VDD pin, VDDUSB included

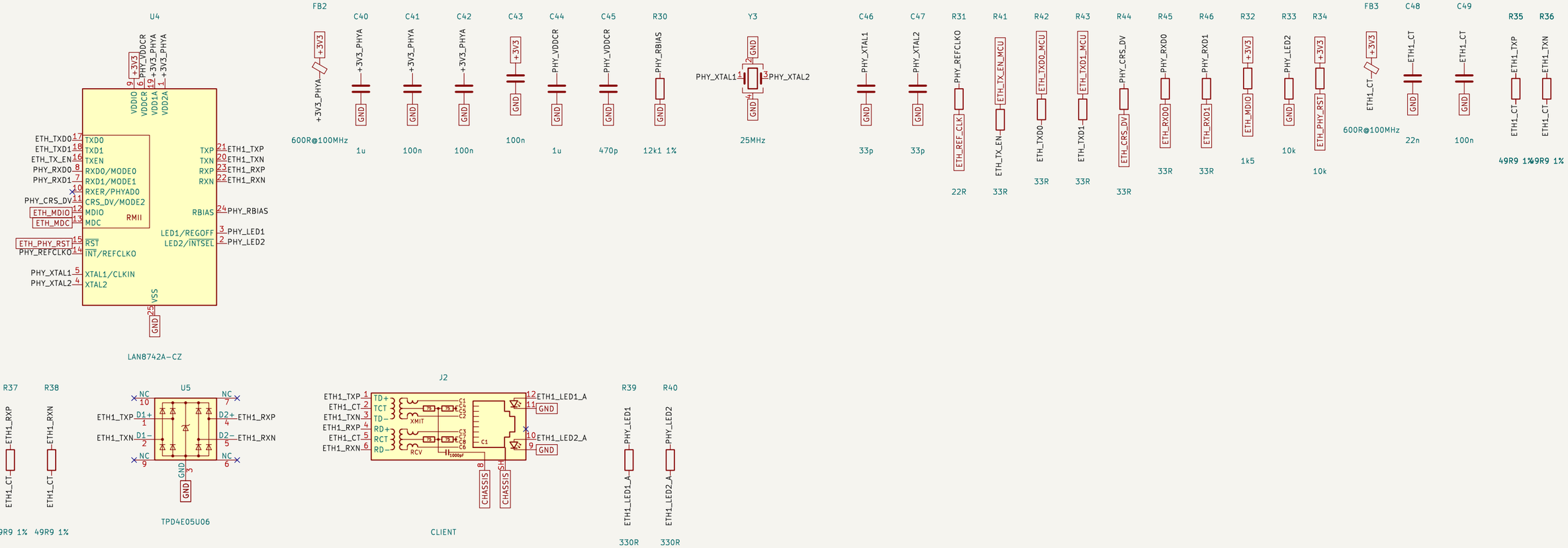
FB1: VDDA / VREF+ filter
C29: VCAP, CHECK value and count against the H5 datasheet
Y1: HSE, CL 10 pF class, CHECK against the DK
C31: CHECK for the crystal's CL
Y2: CL 6 pF, ESR per AN2867
C33: AN2867, CHECK for the crystal
JP1: fit a jumper for the ST bootloader

J6: FTDI TTL-232R-3V3 order: 1 GND, 2 CTS, 3 VCC (not connected), 4 cable TXD, 5 cable RXD, 6 RTS
R23: hardware revision 0: both pulled down
R25: NOT FITTED,
R26: NOT FITTED,
J7: shunt with pull tab; silkscreen 1-2 CORE / 2-3 I/O NODE / none = OFF: reachable only with the housing open
C36: ESD, at J7

Generated from design.py
STM32H573VIT6, clocks, reset, boot, SWD, service UART, LEDs
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Sheet: /MCU/
File: mcu.kicad_sch

Title: CEMS core board: MCU

Size: A2	Date: 2026-10-25	**Rev: 0.1**
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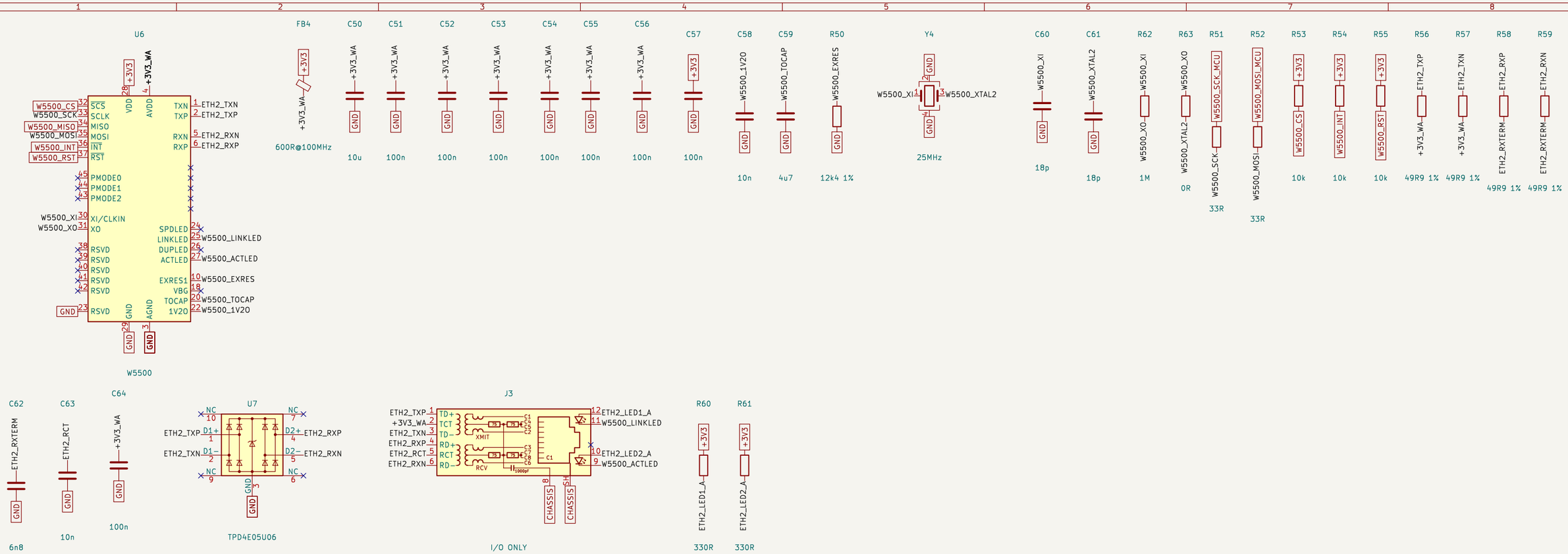
Notes
U4: straps: MODE[2:0] internal pull-ups = all capable; PHYADO internal pull-down = address 0; nINTSEL pulled low (R33) = pin 14 is REFCLKO, 50 MHz to the MCU; REGOFF internal pull-down = regulator on
C41: at pin 1
C42: at pin 19
C43: VDDIO
C44: VDDCR, with C45
Y3: CL 20 pF, +/-50 ppm total (LAN8742A datasheet)
C46: for CL 20 pF with about 4 pF stray
R31: series, at the PHY
R41: series, at the MCU
R42: series, at the MCU
R43: series, at the MCU
R44: series, at the PHY
R45: series, at the PHY
R46: series, at the PHY
R33: strap nINTSEL = 0 (REFCLKO mode)
R34: PHY runs if the firmware leaves PD11 as input
U5: flow-through: route each line across pins 1/10, 2/9, 4/7, 5/6
J2: client network
R39: LED1 = link/activity, active high (REGOFF strap low)
R40: LED2 = speed, active high (nINTSEL strap low)

Generated from design.py
LAN8742A on RMII, client RJ45 (J2)
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Sheet: /Ethernet client port/
File: eth_client.kicad_sch

Title: CEMS core board: Ethernet client port

Size: A3	Date: 2026-10-25	**Rev: 0.1**
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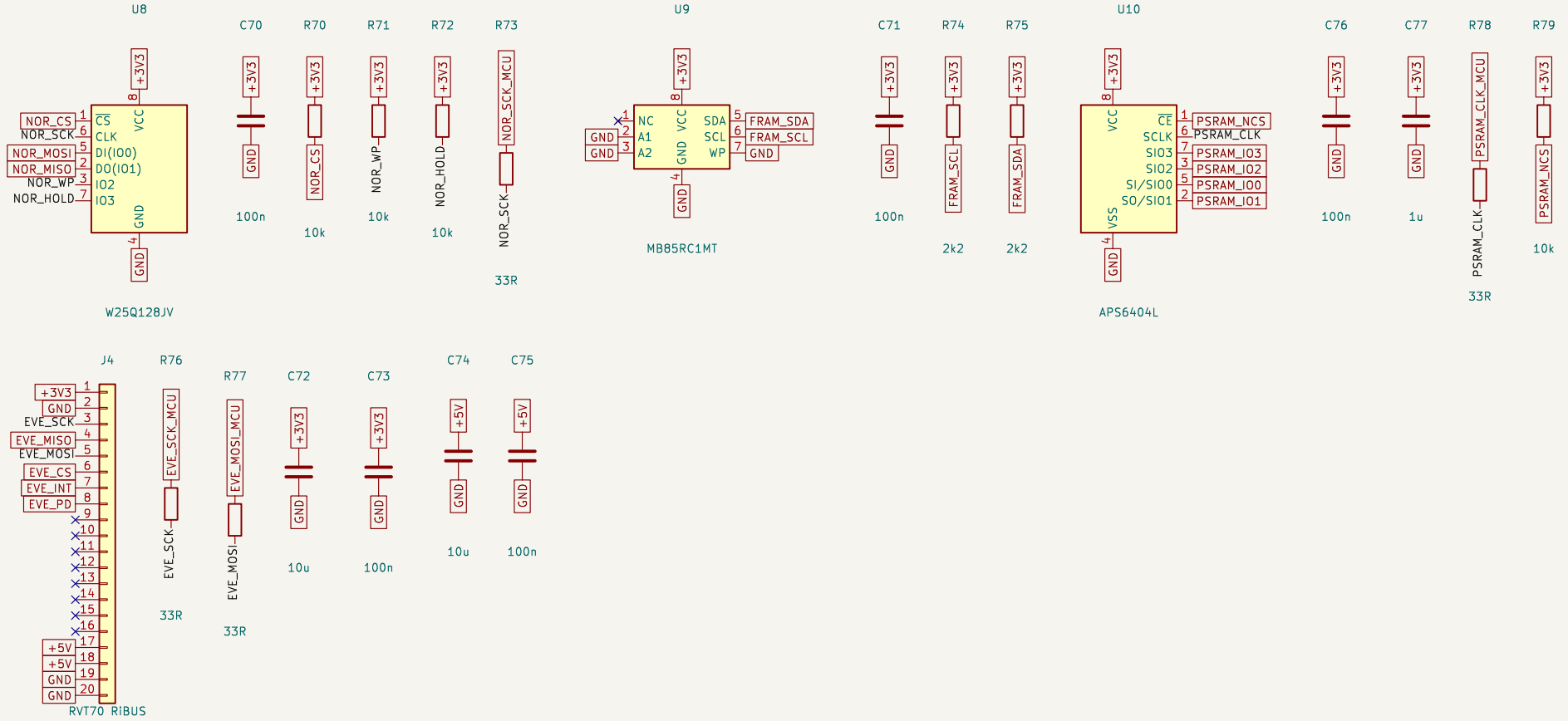


Notes
U6: PMODE[2:0] internal pull-ups = all capable, auto-negotiation; RSVD pin 23 to ground, RSVD 38–42 NC; VBG open. No auto-MDIX: the I/O board's LAN8742A does HP auto-MDIX, so a straight cable works
C51: one per AVDD pin
C52: one per AVDD pin
C53: one per AVDD pin
C54: one per AVDD pin
C55: one per AVDD pin
C56: one per AVDD pin
C57: VDD pin 28
Y4: own crystal, CL 18 pF, +/-30 ppm (W5500 datasheet); separate from the MCU's HSE

C60: W5500 reference circuit
R62: W5500 reference circuit
R63: drive-level option, W5500 reference circuit
R51: series, at the MCU
R52: series, at the MCU
C64: TX centre tap
J3: I/O boards only; silkscreen I/O ONLY
R60: link LED; W5500 LED outputs sink (active low)
R61: activity LED

Generated from design.py
W5500 on SPI4, I/O RJ45 (J3), I/O boards only
EnviroDevelo

Sheet: /Ethernet I/O port/		
File: eth_io.kicad_sch		
Title: CEMS core board: Ethernet I/O port		
Size: A3	Date: 2026-10-25	**Rev: 0.1**
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Notes
U8: event log and data logger (H3)
R73: series, at the MCU
U9: 128 KB FRAM (H2), I2C address 0x50. A1 = A2 = 0, WP low = writes enabled (pinout per datasheet)
U10: 8 MB QSPI PSRAM on OCTOSPI1, up to about 100 MHz: keep it next to the MCU, traces under 30 mm and roughly matched
C76: at U10 VDD
R78: series, at the MCU
R79: keeps the PSRAM deselected through reset
J4: RiBUS pinout from the RVT70HSBNWC00 V1.1A datasheet; 9–12 (GPIO, audio, QSPI) left open; 13–16 NC. CHECK contact side against the cable at mechanical layout
R76: series, at the MCU
R77: series, at the MCU
C72: at J4
C73: at J4
C74: at J4, backlight
C75: at J4

Generated from design.py
NOR flash, FRAM, RVT70 display FFC (J4)
EnviroDevelo

Sheet: /Memories and display/
File: periph.kicad_sch

Title: CEMS core board: Memories and display

Size: A4	Date: 2026-10-25	**Rev: 0.1**
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